

Theorem 1. *Let $a \geq 1$ and $b > 1$ be constants, let $f(n)$ be a function, and let $T(n)$ be defined on the non-negative numbers by the recurrence*

$$T(n) = aT(n/b) + f(n) \tag{1}$$

where we interpret n/b to mean either $\lfloor n/b \rfloor$ or $\lceil n/b \rceil$. Then $T(n)$ has the following asymptotic bounds:

1. *If $f(n) = \mathcal{O}(n^{\epsilon} \log n)$ for some constant $\epsilon > 0$, then $T(n) = \Theta(n^{\epsilon} \log n)$.*
2. *If $f(n) = \Theta(n^{\epsilon} \log n)$, then $T(n) = \Theta(n^{\epsilon} \log n)$.*
3. *If $f(n) = \Omega(n^{\epsilon} \log n)$ for some constant $\epsilon > 0$, and if*

$$af(n/b) < cf(n)$$

for some constant $c < 1$ and sufficiently large n , then $T(n) = \Theta(n^{\epsilon} \log n)$.

Question #1

Problem #1.1

From Equation 1, we have the following:

$$a = 2 \quad b = 4 \quad f(n) = 1$$

From this,

$$\begin{aligned} f(n) &= \mathcal{O}(n^a) \\ 1 &= \mathcal{O}(n^a) \\ &= \mathcal{O}(n^4) \\ &= \mathcal{O}(n^4) \\ &= \mathcal{O}(n^4) \end{aligned}$$

Therefore, the tight asymptotic bound for the recurrence is

$$\begin{aligned} T(n) &\in \Theta(n^a) \\ &\in \Theta(n^4) \\ &\in \Theta(n^4) \\ &\in \Theta(n^4) \end{aligned}$$

Therefore, the bound is $\Theta(n^4)$.

Problem #1.2

From Equation 1, we have the following:

$$a = 2 \quad b = 4 \quad f(n) = \sqrt{n}$$

From this,

$$\sqrt{n} = \Theta(n^{1/2})$$

Therefore, the tight asymptotic bound for the recurrence is

$$T(n) \in \Theta(n^{1/2})$$

Problem #1.3

From Equation 1, we have the following:

$$a = 2 \quad b = 4 \quad f(n) = n$$

From this,

$$n = \Omega(n \log m_1)$$

Therefore, the tight asymptotic bound for the recurrence is

$$T(n) \in \Theta(n \log m_1)$$

Problem #1.4

From Equation 1, we have the following:

$$a = 2 \quad b = 4 \quad f(n) = n^2$$

From this,

$$n^2 = \Omega(n \log n)$$

Therefore, the tight asymptotic bound for the recurrence is

$$T(n) \in \Theta(n \log n)$$

Question #2

Recall Strassen's algorithm belongs to the complexity class $\Theta(n^{\lg 7})$. By using the Case 1 of the Main Theorem, we must find a that solves the inequality $\log_4 a < \lg 7$. The integer that solves the equation $\log_4 a = \lg 7$ is 49; therefore,

$$a = 48$$

Question #3

Because the process is identical to Problem , the work is omitted; only answers are provided.

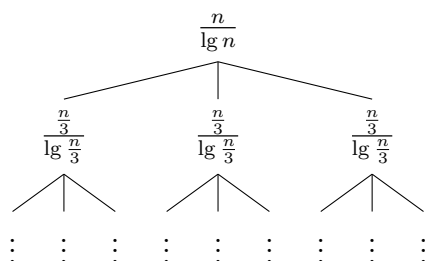
1. Main Theorem (*Case #3*), $T(n) \in \Theta(n \log n)$
2. Main Theorem (*Case #3*), $T(n) \in \Theta(n \log n)$
3. Main Theorem (*Case #2*), $T(n) \in \Theta(n \log n)$
4. Main Theorem (*Case #3*), $T(n) \in \Theta(n \log n)$
5. Main Theorem (*Case #1*), $T(n) \in \Theta(n \log n)$
6. Main Theorem (*Case #2*), $T(n) \in \Theta(n \log n)$
7. Because $(n - 2) \in \mathcal{O}(n \log n)$, we can write an equivalent recurrence relation,

$$T(n) = T(n/2) + n^2$$

Because $\log_1 1$ is undefined, we cannot use the Main Theorem; but we do not need it. We have n^2 work to do n times. This implies that $T(n) \in \Theta(n^3)$.

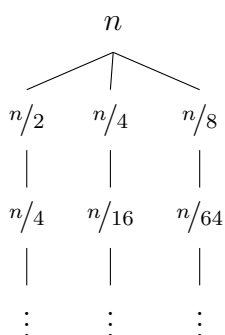
Question #4

1. Main Theorem (*Case #1*), $T(n) \in \Theta(n \lg n)$
2. Using a recursion tree,



We see that the $T(n) \in \Theta(n \lg n)$.

3. Main Theorem (*Case #3*), $T(n) \in \Theta(n \lg n)$.
4. We augment the equation to be $3T(n/3) + n/2$, where we can use Main Theorem (*Case #2*), $T(n) \in \Theta(n \lg n)$.
5. With similar reasoning as in Step #2, we use a recursion tree to get the results $T(n) \in \Theta(n \lg n)$
6. We solve so via the substitution method. We use a recursion tree to get our initial guess.



Guessing $T(n) \in \Theta(n \lg n)$, we get the following:

$$\begin{aligned}
 T(n) &= T(n/2) + T(n/4) + T(n/8) + n \\
 &= c^{n/2} + c^{n/4} + c^{n/8} + n \\
 &= \frac{7}{8}n + n \\
 &\leq cn
 \end{aligned}$$

We see this inequality holds for $c \geq 8$. Therefore, $T(n) \in \Theta(n \lg n)$.

7. Using a recursion tree,

$$\begin{array}{c} \frac{1}{n} \\ | \\ \frac{1}{n-1} \\ | \\ \frac{1}{n-2} \\ | \\ \vdots \end{array}$$

We see that $T(n) \in \Theta(n \lg n)$.

8. Using a recursion tree,

$$\begin{array}{c} \lg n \\ | \\ \lg(n-1) \\ | \\ \lg(n-2) \\ | \\ \vdots \end{array}$$

We see that $T(n) \in \Theta(n \lg n)$.

9. Using a recursion tree,

$$\begin{array}{c} \frac{1}{\lg n} \\ | \\ \frac{1}{\lg(n-2)} \\ | \\ \frac{1}{\lg(n-4)} \\ | \\ \vdots \end{array}$$

We see that $T(n) \in \Theta(n \lg n)$.

10. We guess

$$T(n) \in \Theta(n \lg n)$$

$$\begin{aligned}
T(n) &= \sqrt{n} T(\sqrt{n}) + n \\
&= \sqrt{n} T(c\sqrt{n} \lg \lg \sqrt{n}) + n \\
&= c n \lg \lg n - c n + n \\
&\leq c n \lg \lg n
\end{aligned}$$

We see this inequality for $c \in \mathbb{R}^+$. Therefore $T(n) \in \Theta(n \lg \lg n)$.

Question #5

Options *c* and *e*.

Question #6

1. We choose a root among the vertices, call it v_0 . If we choose the k th smallest element, the smaller subtree will have $i - 1$ vertices and the larger will have $n - i$ vertices. Summing over all possibilities, we get the following form for b_n :

$$\begin{aligned} b_n &= \sum_{k=1}^n b_{k-1} b_{n-k} \\ &= \sum_{k=0}^{n-1} b_k b_{n-k-1} \end{aligned}$$

- 2.

$$\begin{aligned} B(x) &= \sum_{n=0}^{\infty} b_n x^n \\ &= 1 + \sum_{n=1}^{\infty} b_n x^n \\ &= 1 + \sum_{n=1}^{\infty} \sum_{k=0}^{n-1} b_k b_{n-k-1} x^{n-k-1} \\ &= 1 + x \sum_{n=1}^{\infty} \sum_{k=0}^{n-1} b_k x^k b_{n-k-1} x^{n-k-1} \\ &= 1 + x \sum_{n=0}^{\infty} \sum_{k=0}^n b_k x^k b_{n-k} x^{n-k} \\ &= 1 + x B(x)^2 \end{aligned}$$

3. We use the Taylor expansion of $\sqrt{1-4x}$, we have the following

$$\begin{aligned} B(x) &= \frac{1}{2x} \left(1 - \sum_{n=0}^{\infty} \frac{1}{1-2n} \binom{2n}{n} x^n \right) \\ &= -\frac{1}{2x} \sum_{n=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{1-2n} \binom{2n}{n} x^n \\ &= \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{2n-1} \binom{2n}{n} x^{n-1} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2n+1} \binom{2n+2}{n+1} x^n \end{aligned}$$

Therefore $b_n = \frac{1}{n+1} \binom{2n}{n}$.

4.

$$\begin{aligned}
b_n &= \frac{1}{n+1} \frac{(2n)!}{n!n!} \\
&\approx \frac{1}{n+1} \frac{\sqrt{4\pi n} (2n/e)^{2n}}{2\pi n (n/e)^{2n}} \\
&= \frac{1}{n+1} \frac{4^n}{\sqrt{\pi n}} \\
&= \left(\frac{1}{n} + \left(\frac{1}{n+1} - \frac{1}{n} \right) \right) \frac{4^n}{\sqrt{\pi n}} \\
&= \left(\frac{1}{n} - \frac{1}{n^2 + n} \right) \frac{4^n}{\sqrt{\pi n}} \\
&= \frac{1}{n} \left(1 - \frac{1}{n+1} \right) \frac{4^n}{\sqrt{\pi n}} \\
&= \frac{4^n}{\sqrt{\pi n}^{3/2}} (1 + O(1/n))
\end{aligned}$$